Faculty of Engineering & Technology

Electrical & Computer Engineering Department

COMMUNICATION SYSTEM – ENEE3309

Assignment

Fourier series and single side band

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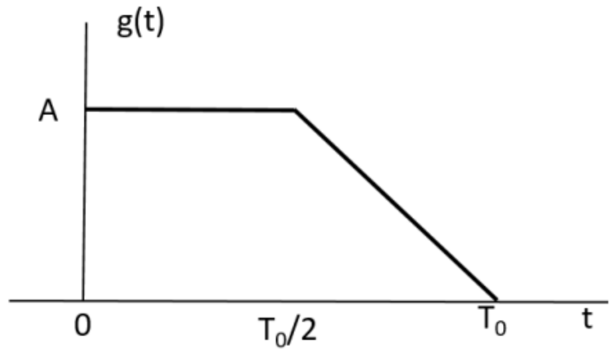
Section: 2

Date:14/1/2023

Contents

Section 0:

Consider the periodic signal g(t), for which one period is shown in the figure below



One Period of the periodic signal g(t)

Plotting one period of g(t)

```
%If we take the period T0 =4 & A =1 The figure will be %

syms t;
%----------------------------------g(t)--------------------------------
T0=4; % assume that the period here is 4.
A=1;
y1 = @(t)heaviside(t-(T0/2));
y2= @(t)A*triangularPulse((2/T0)*(t-(T0/2)));
y3 =@(t) A*rectangularPulse((2/T0)*(t-(T0/4)));


f = @(t) y3(t)+ y1(t)*y2(t);


fplot(f(t),[0,6])
```

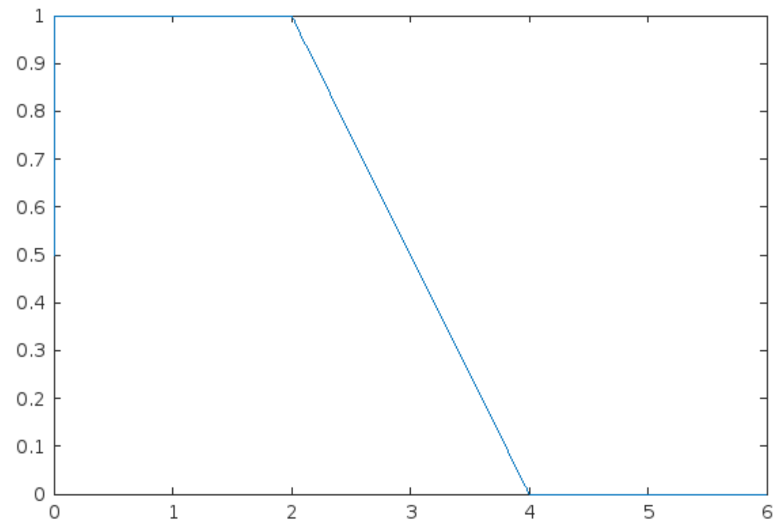
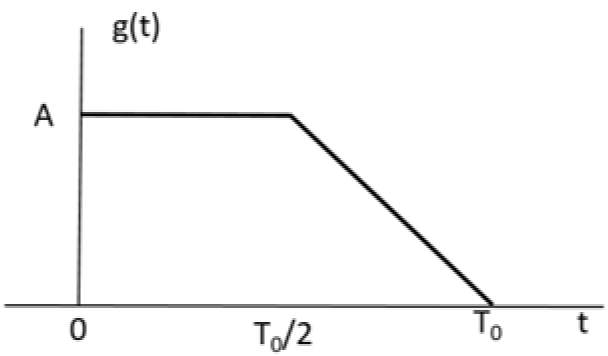


Figure 1:figure of one period of g(t)



If we notice the function, it's a0 = A (3/4), a0 doesn't depend on the value of T0.

$$a0 = \frac{1}{T0}\left(\int G(t)\right) \qquad a0 = \frac{1}{T0}(area\ rectangle\ +\ areatriangle\ )$$

$$a0 = \frac{1}{T0}\left(A\left(\frac{T0}{2}\right) + \frac{1}{2}\left(T0 - \frac{T0}{2}\right)A\right)$$

$$a0 = \left(A\left(\frac{1}{2}\right) + \frac{1}{2}\left(\frac{1}{2}\right)A\right) \qquad a0 = \left(\frac{3}{4}A\right)$$

$$g(t) = a_0 + \sum_{n=1}^{\infty} (a_n \cos n\omega_0 t + b_n \sin n\omega_0 t)$$

Now, consider the approximate signal:

$$g_a(t) = a_0 + \sum_{n=1}^{K} (a_n \cos n\omega_0 t + b_n \sin n\omega_0 t)$$

1. **Find $a_0, a_1, a_2, a_3, b_1, b_2, and\ b_3$ (you can use matlab or any other code to find numerical values of the coefficients)**

Finding the numerical value of the coefficients

Section 1

```
syms t n;
%-------------------------------------
n= 1:5; T0=0.1; f0=1/T0; A=1; w0 =2*pi*f0;




y1 = @(t)heaviside(t-(T0/2));
y2= @(t)A*triangularPulse((2/T0)*(t-(T0/2)));
y3 =@(t) A*rectangularPulse((2/T0)*(t-(T0/4)));
f = @(t) y3(t)+ y1(t)*y2(t);


a0 =(1/T0)*( int(f,t,0,T0))
a0 =
```

$$\frac{3}{4}$$

```
an =(2/T0)*(int(f*cos(n*w0*t),t,0,T0))

an =
```

$$\left(-\frac{2}{\pi^2} \quad 0 \quad -\frac{2}{9\,\pi^2} \quad 0 \quad -\frac{2}{25\,\pi^2}\right)$$

```
bn =(2/T0) *(int(f*sin(n*w0*t), t,0, T0))

bn =
```

$$\left(\frac{1}{\pi} \quad \frac{1}{2\,\pi} \quad \frac{1}{3\,\pi} \quad \frac{1}{4\,\pi} \quad \frac{1}{5\,\pi}\right)$$

a0 =

$$\frac{3}{4}$$

an =

$$\begin{array}{c} \begin{array}{ccccc} a1 & a2 & a3 & a4 & a5 \end{array} \\ \left(\begin{array}{ccccc} -\frac{2}{\pi^2} & 0 & -\frac{2}{9\pi^2} & 0 & -\frac{2}{25\pi^2} \end{array} \right) \end{array}$$

bn =

$$\begin{array}{c} \left(\begin{array}{ccccc} \frac{1}{\pi} & \frac{1}{2\pi} & \frac{1}{3\pi} & \frac{1}{4\pi} & \frac{1}{5\pi} \end{array} \right) \\ \begin{array}{ccccc} b1 & b2 & b3 & b4 & b5 \end{array} \end{array}$$

Figure 2:numerical value of the coefficients

2. **Use matlab to plot $g(t)$ and $g_a(t)$ for K = 3, on the same figure for one cycle of $g(t)$.**

Plotting G(t) for K =30 which nearly indicates for K= infinity and plotting Ga(t) For K =3.

Each G(t) and Ga(t) and G(t) &Ga(t) have its plot alone.

Section 2

```
syms t n;
%-------------------------------------
T0=0.1; f0=1/T0; A=1; w0 =2*pi*f0; sum =0; sum2 =0;
y1 = @(t)heaviside(t-(T0/2));
y2= @(t)A*triangularPulse((2/T0)*(t-(T0/2)));
y3 =@(t) A*rectangularPulse((2/T0)*(t-(T0/4)));
f = @(t) y3(t)+ y1(t)*y2(t);

fplot(f(t),[0,1]) % plot of g(t) one period

a0 =(1/T0)*( int(f,t,0,T0))
```

a0 =

$$\frac{3}{4}$$

```
%the Ga(t)  with k =3

    for n= 1 :1:3 % Summation from 1 to 3

        an =(2/T0)*(int(f*cos(n*w0*t),t,0,T0)) ;
        bn =(2/T0)*(int(f*sin(n*w0*t),t,0,T0)) ;
        x= an*cos(n*w0*t)+ bn*sin(n*w0*t);
```

```
        sum = sum +x ;

    end % for n loop

    g(t)= a0 +sum
```

g(t) =

$$\frac{\sin(20\,\pi\,t)}{\pi} - \frac{2\cos(60\,\pi\,t)}{9\,\pi^2} - \frac{2\cos(20\,\pi\,t)}{\pi^2} + \frac{\sin(40\,\pi\,t)}{2\,\pi} + \frac{\sin(60\,\pi\,t)}{3\,\pi} + \frac{3}{4}$$

```
    hold on

        fplot(g,[0,0.6],'r')

    title("Both G(t) & Ga(t)")
```

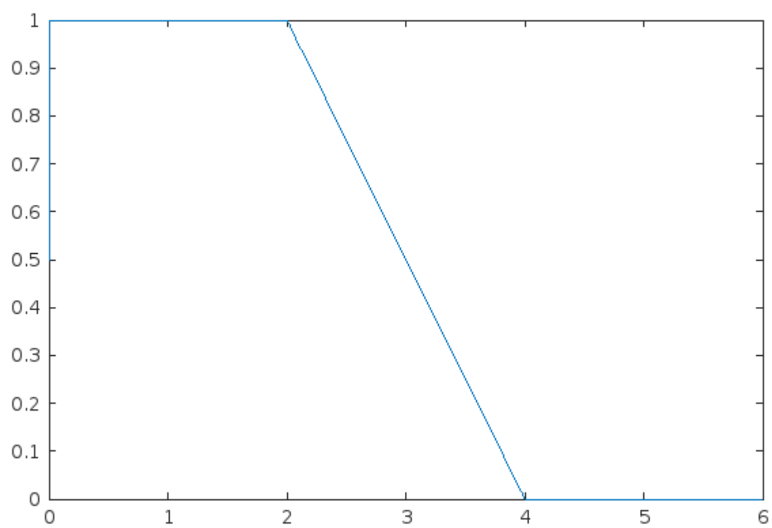


Figure 2: figure of one period of g(t

```
title("Ga(t)")
```

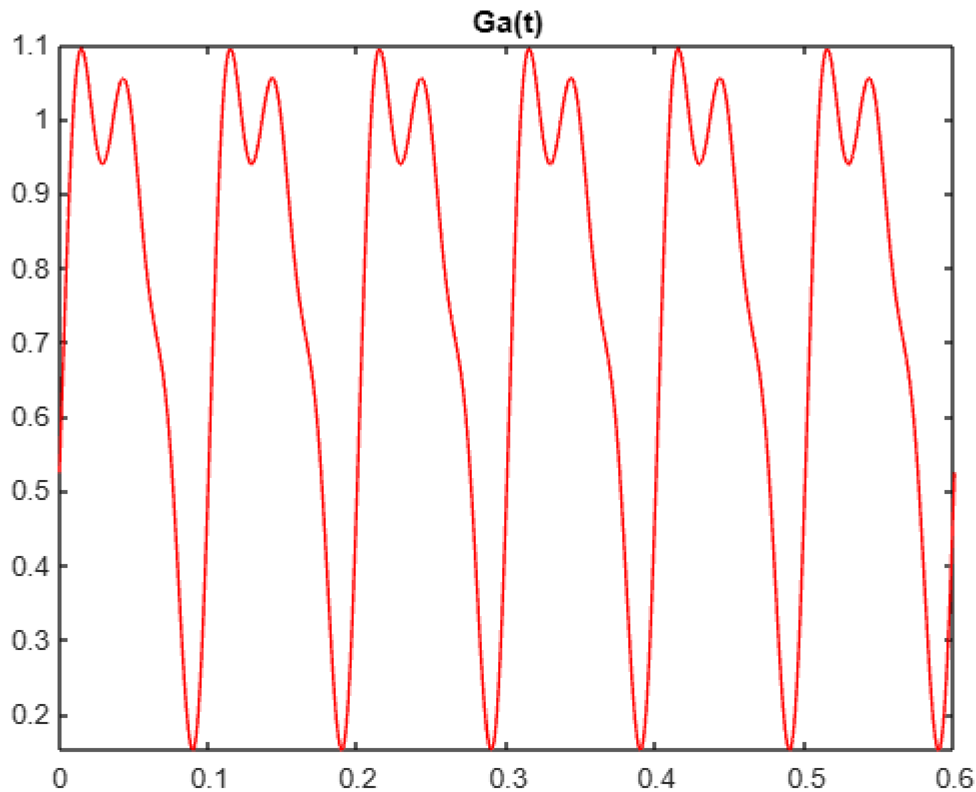


Figure 3:Ga(t) plot when k =3

```
t)")
```

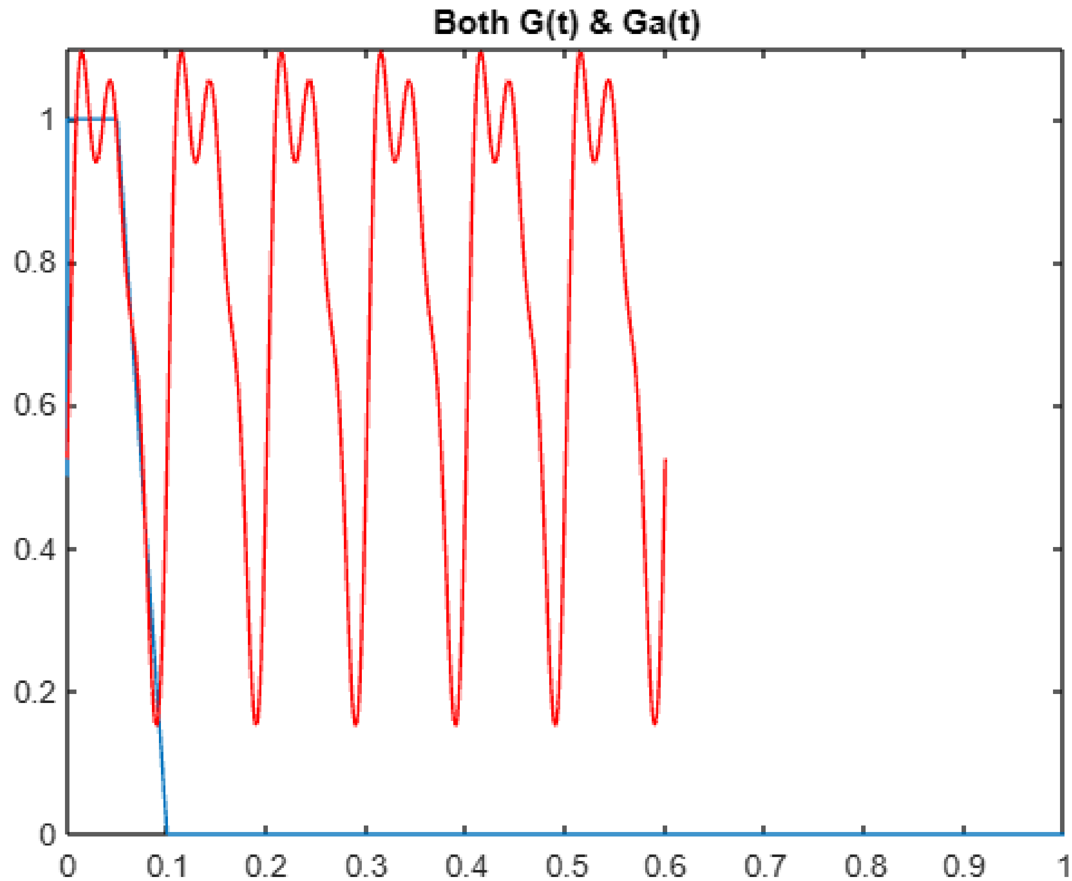


Figure 4:Both Ga(t)&G(t) on the same figure

Another way of understanding If we choose that g(t) is Fourier series k = infinity

```
for n= 1 :1:30 % Summation from 1 to 30


        an =(2/T0)*(int(f*cos(n*w0*t),t,0,T0)) ;
        bn =(2/T0)*(int(f*sin(n*w0*t),t,0,T0)) ;
        x2= an*cos(n*w0*t)+ bn*sin(n*w0*t);
        sum2 = sum2 +x2 ;

        end % for n loop


    g2(t)= a0 +sum2
g2(t) =
```

$$\frac{\sin(20\pi t)}{\pi} - \frac{2\cos(60\pi t)}{9\pi^2} - \frac{2\cos(100\pi t)}{25\pi^2} - \frac{2\cos(140\pi t)}{49\pi^2} - \frac{2\cos(180\pi t)}{81\pi^2} - \frac{2\cos(220\pi t)}{121\pi^2} - \frac{2\cos(260\pi t)}{169\pi^2} - \frac{2\cos(300\pi t)}{225\pi^2} - \frac{2\cos(340\pi t)}{289\pi^2} - \frac{2\cos(380\pi t)}{361\pi^2} - \frac{2\cos(420\pi t)}{441\pi^2} - \frac{2\cos(460\pi t)}{529\pi^2} - \frac{2\cos(500\pi t)}{625\pi^2} - \frac{2\cos(540\pi t)}{729\pi^2} - \frac{2\cos(580\pi t)}{841\pi^2} - \frac{2\cos(20\pi t)}{\pi^2} + \frac{\sin(40\pi t)}{2\pi} + \frac{\sin(60\pi t)}{3\pi} + \frac{\sin(80\pi t)}{4\pi} + \frac{\sin(100\pi t)}{5\pi} + \frac{\sin(120\pi t)}{6\pi} + \frac{\sin(140\pi t)}{7\pi} + \frac{\sin(160\pi t)}{8\pi} + \frac{\sin(180\pi t)}{9\pi} + \sin$$

```
            fplot(g2,[0,0.6],'g')



    title(" G(t) ")
```

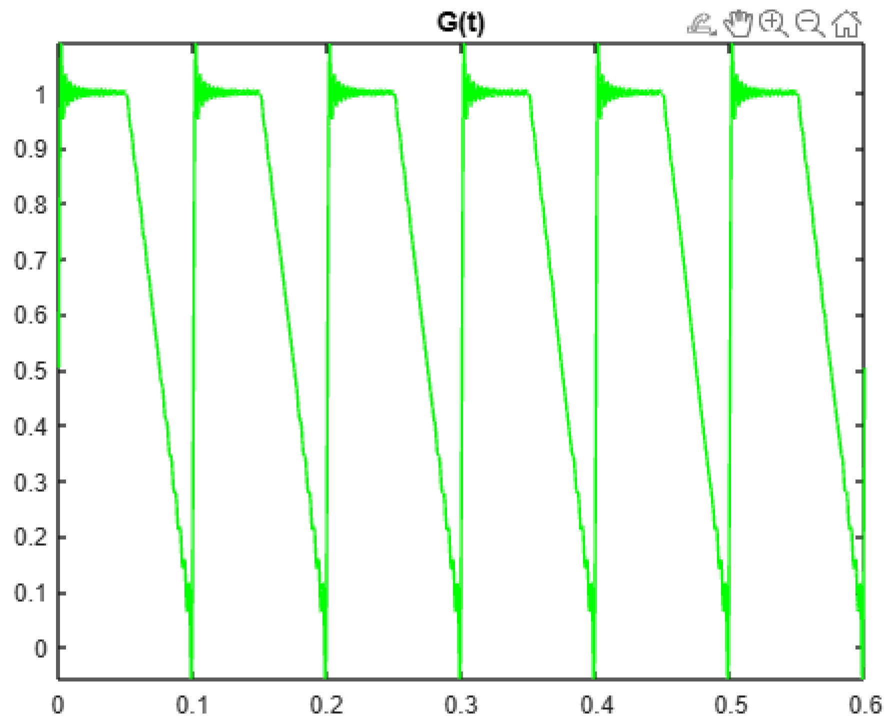
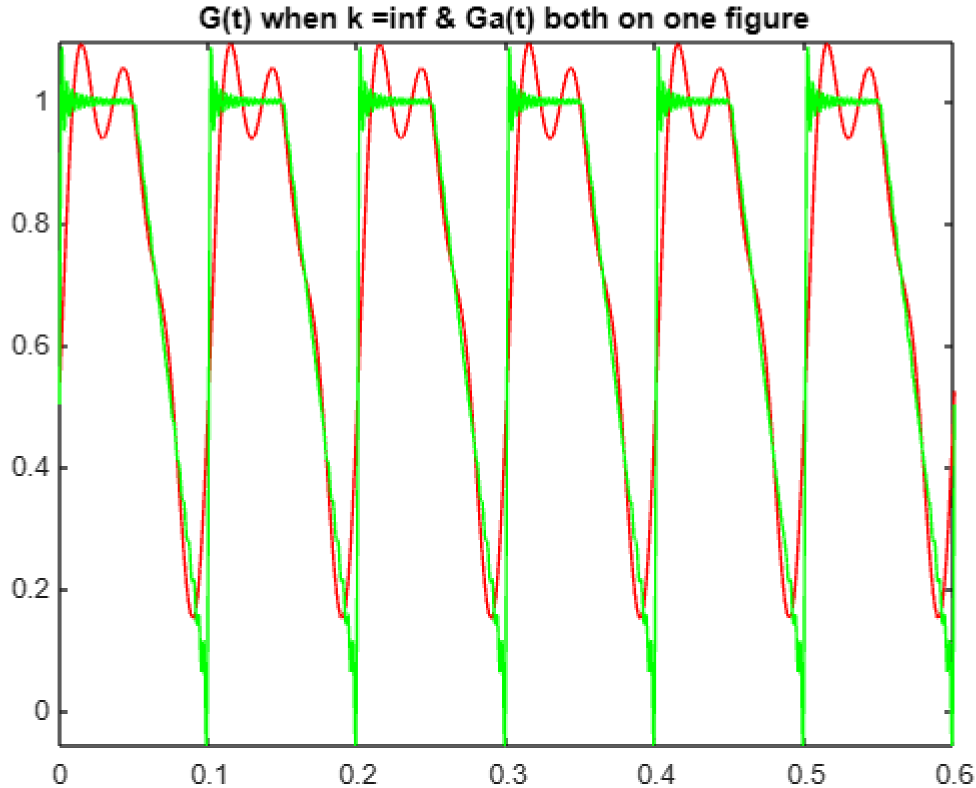


Figure 5: g(t) when k =30, aproxmilty inf

```
G(t) when k =inf & Ga(t) both on one figure
```


G(t) when k =inf & Ga(t) both on one figure
1
0.8
0.6
0.4
0.2
0
0
0.1
0.2
0.3
0.4
0.5
0.6

Section 3

3. The mean square error between $g(t)$ and $g_a(t)$ is defined as

$$MSE = \frac{1}{T_0}\left(\int_0^{T_0} \left(g(t) - g_a(t)\right)^2 dt\right)$$

Find the mean square error for K=1, 2, and 3. Summarize your results in a table.

Finding the mean square error for K=1 ,2,3.

Section 3

```
syms t n;
%-----------------------------------
T0=0.1; f0=1/T0; A=1; w0 =2*pi*f0; sum =0; sum2 =0;
y1 = @(t)heaviside(t-(T0/2));
y2= @(t)A*triangularPulse((2/T0)*(t-(T0/2)));
y3 =@(t) A*rectangularPulse((2/T0)*(t-(T0/4)));
f = @(t) y3(t)+ y1(t)*y2(t);
a0 =(1/T0)*( int(f,t,0,T0))
```

a0 =

$\frac{3}{4}$

```
%the Ga(t) with k =3
 for n= 1 :1:3 % Summation from 1 to 3
 an =(2/T0)*(int(f*cos(n*w0*t),t,0,T0)) ;
 bn =(2/T0)*(int(f*sin(n*w0*t),t,0,T0)) ;
 x= an*cos(n*w0*t)+ bn*sin(n*w0*t);
 if(n==1)
 gk1= sum +x +a0 % ga when k =1

 elseif(n==2)
 gk2= sum +x+ a0  % ga when k=2

 elseif(n==3)
 gk3= sum +x +a0  % ga when k=2
 end
 sum = sum +x ;

 end % for n loop
```

gk1 =

$$\frac{\sin(20\,\pi\,t)}{\pi} - \frac{2\cos(20\,\pi\,t)}{\pi^2} + \frac{3}{4}$$

gk2 =

$$\frac{\sin(20\,\pi\,t)}{\pi} - \frac{2\cos(20\,\pi\,t)}{\pi^2} + \frac{\sin(40\,\pi\,t)}{2\,\pi} + \frac{3}{4}$$

gk3 =

$$\frac{\sin(20\,\pi\,t)}{\pi} - \frac{2\cos(60\,\pi\,t)}{9\,\pi^2} - \frac{2\cos(20\,\pi\,t)}{\pi^2} + \frac{\sin(40\,\pi\,t)}{2\,\pi} + \frac{\sin(60\,\pi\,t)}{3\,\pi} + \frac{3}{4}$$

```
MGk1= (f-gk1)^2;
MGk2= (f-gk2)^2;
MGk3= (f-gk3)^2;

MSE1= (1/T0)*(int(MGk1,t,0,T0))
```

MSE1 =

$$\frac{5}{48} - \frac{10\left(\frac{\pi^2}{20} + \frac{1}{5}\right)}{\pi^4}$$

```
var=vpa(MSE1)
```

var =

0.032974110336129110566421368527565

```
MSE2= (1/T0)*(int(MGk2,t,0,T0))
```

MSE2 =

$$\frac{5}{48} - \frac{10\left(\frac{\pi^2}{16} + \frac{1}{5}\right)}{\pi^4}$$

```
var2 = vpa(MSE2) % k=2
```

var2 =

0.020308962380836889135936435626 35

```
MSE3= (1/T0)*(int(MGk3,t,0,T0))
```

MSE3 =

$$\frac{5}{48} - \frac{10\left(\frac{49\,\pi^2}{720} + \frac{82}{405}\right)}{\pi^4}$$

```
var3 = vpa(MSE3) % k =3
```

var3 =

0.014426526690714807137470223885888

K=1	K=2	K=3
MSE =0.032974110336129110566421368527565 If we take the function Is g(t) on one period	MSE=0.02030896238083688913593643562635	MSE = 0.014426526690714807137470223885888
-	-	-

Another way I understand this section, that you want g(t) is a from k=1 to K=inf for in fourire series formula

I solved another solution to be sure.

Finding the mean square error for K=1 ,2,3.

Section 3 other way of understanding

Section 3

```
syms t n;
%------------------------------------
T0=0.1; f0=1/T0; A=1; w0 =2*pi*f0; sum =0; sum2 =0;


y1 = @(t)heaviside(t-(T0/2));
y2= @(t)A*triangularPulse((2/T0)*(t-(T0/2)));
y3 =@(t) A*rectangularPulse((2/T0)*(t-(T0/4)));
f = @(t) y3(t)+ y1(t)*y2(t);


a0 =(1/T0)*( int(f,t,0,T0))
```

a0 =

$$\frac{3}{4}$$

```
%the Ga(t)  with k =3


    for n= 1 :1:3 % Summation from 1 to 3


        an =(2/T0)*(int(f*cos(n*w0*t),t,0,T0)) ;
        bn =(2/T0)*(int(f*sin(n*w0*t),t,0,T0)) ;
        x= an*cos(n*w0*t)+ bn*sin(n*w0*t);


        if(n==1)
        g1= sum +x +a0; % ga when k =1

        elseif(n==2)
        g2= sum +x+ a0 ; % ga when k=2

        elseif(n==3)
        g3= sum +x +a0 ; % ga when k=2
        end
```

```
        sum = sum +x ;

    end % for n loop

%the g(t) with k= inft

    for n= 1 :1:10 % Summation from 1 to 10, here we choose 10 because

        an =(2/T0)*(int(f*cos(n*w0*t),t,0,T0)) ;
        bn =(2/T0)*(int(f*sin(n*w0*t),t,0,T0)) ;
        x= an*cos(n*w0*t)+ bn*sin(n*w0*t);

        sum = sum +x ;

    end % for n loop

    gk =sum+a0;

          MGk1= (gk-g1)^2;
          MGk2= (gk-g2)^2;
          MGk3= (gk-g3)^2;

          MSE1= (1/T0)*(int(MGk1,t,0,T0))
```

```
MSE1 =
```

$$\frac{10\left(\frac{3344249\,\pi^2}{25401600}+\frac{10353157336}{49228003125}\right)}{\pi^4}$$

```
        var1 = vpa(MSE1)%k =1
var1 =
```

0.15498487889611552007747654188894

```
        MSE2= (1/T0)*(int(MGk2,t,0,T0))
MSE2 =
```

$$\frac{10\left(\frac{2391689\,\pi^2}{25401600}+\frac{10353157336}{49228003125}\right)}{\pi^4}$$

```
        var2 = vpa(MSE2) % k=2
var2 =
```

0.11698943503023885578602174318529

```
        MSE3= (1/T0)*(int(MGk3,t,0,T0))
MSE3 =
```

$$\frac{11\ (88776111375\,\pi^2+232459763456)}{1260236880000\,\pi^4}$$

```
        var3 = vpa(MSE3) % k =3
var3 =
```

0.099342127959872609790623107963881

K=1	K=2	K=3
MSE =0.15498487889611552007747654188894 If we take the function Is g(t) on k=inf	MSE=0.11698943503023885578602174318529	MSE = 0.099342127959872609790623107963881

Section 4

4. **If $g_a(t)$ (when K = 3) is multiplied by the carrier $c(t) = 10\ \cos 2\pi(200)t$ followed by an ideal bandpass filter to generate the single sideband signal $s(t)$, find s(t) and its spectrum.**

Section 4

```
syms t;


%-------------------------------------
T0=0.1; f0=1/T0; A=1; w0 =2*pi*f0; sum =0; sum2 =0;
fs=1000;
z=0:1/fs:0.5;




y1 = @(t)heaviside(t-(T0/2));
y2= @(t)A*triangularPulse((2/T0)*(t-(T0/2)));
y3 =@(t) A*rectangularPulse((2/T0)*(t-(T0/4)));
f = @(t) y3(t)+ y1(t)*y2(t);
c=  10*cos(2*pi*(200)*t);


a0 =(2/T0)*( int(f,t,0,T0))
```

a0 =

$\frac{3}{2}$

```
%the Ga(t)  with k =3


    for n= 1:1:3


        an =(2/T0)*(int(f*cos(n*w0*t),t,0,T0)) ;
        bn =(2/T0)*(int(f*sin(n*w0*t),t,0,T0)) ;
        x= an*cos(n*w0*t)+ bn*sin(n*w0*t);
        sum = sum +x ;
```

```
end % for n loop

ga= a0 +sum;

Sdoubleside(t)=  ga*(c)
```

Sdoubleside(t) =

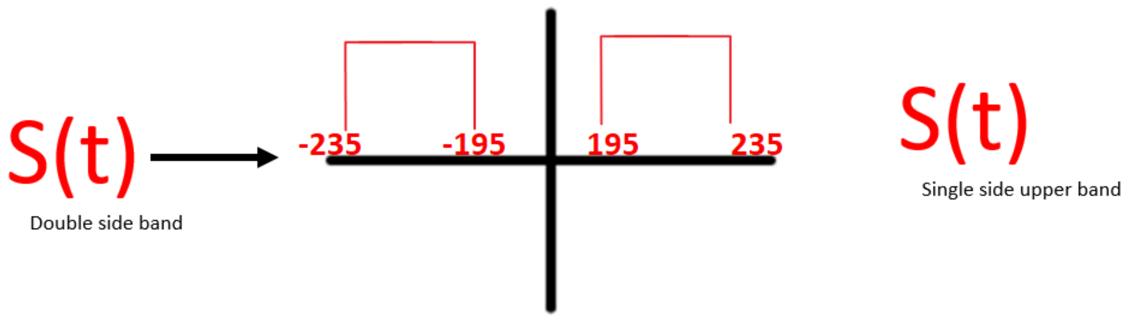
$$10\cos(400\pi t)\left(\frac{\sin(20\pi t)}{\pi}-\frac{2\cos(60\pi t)}{9\pi^2}-\frac{2\cos(20\pi t)}{\pi^2}+\frac{\sin(40\pi t)}{2\pi}+\frac{\sin(60\pi t)}{3\pi}+\frac{3}{2}\right)$$

Now Fourier transform for F[**Sdoubleside(t)**] =

$$\mathcal{F}_t\left[10\cos(400\pi t)\left(\frac{\sin(20\pi t)}{\pi}-\frac{2\cos(60\pi t)}{9\pi^2}-\frac{2\cos(20\pi t)}{\pi^2}+\frac{\sin(40\pi t)}{2\pi}+\frac{\sin(60\pi t)}{3\pi}+\frac{3}{2}\right)\right](f)$$

$$\begin{aligned}
= \frac{1}{36\pi^2}\, 5\big(&(-4-6 i\pi)\,\delta(f-230) - 9 i\pi\,\delta(f-220) - 18 i\pi\,\delta(f-210) - \\
&36\,\delta(f-210) + 54\pi^2\,\delta(f-200) + 18 i\pi\,\delta(f-190) - \\
&36\,\delta(f-190) + 9 i\pi\,\delta(f-180) + 6 i\pi\,\delta(f-170) - 4\,\delta(f-170) - \\
&6 i\pi\,\delta(f+170) - 4\,\delta(f+170) - 9 i\pi\,\delta(f+180) - 18 i\pi\,\delta(f+190) - \\
&36\,\delta(f+190) + 54\pi^2\,\delta(f+200) + 18 i\pi\,\delta(f+210) - \\
&36\,\delta(f+210) + 9 i\pi\,\delta(f+220) + 6 i\pi\,\delta(f+230) - 4\,\delta(f+230)\big)
\end{aligned}$$

After Pass band filter to get the upper band



S(f)single side =

$$= \frac{1}{36\pi^2} 5\big((-4 - 6 i \pi)\, \delta(f - 230) - 9 i \pi\, \delta(f - 220) - 18 i \pi\, \delta(f - 210) - 36\, \delta(f - 210) + 54 \pi^2\, \delta(f - 200) + 54 \pi^2\, \delta(f + 200) + 18 i \pi\, \delta(f + 210) - 36\, \delta(f + 210) + 9 i \pi\, \delta(f + 220) + 6 i \pi\, \delta(f + 230) - 4\, \delta(f + 230)\big)$$

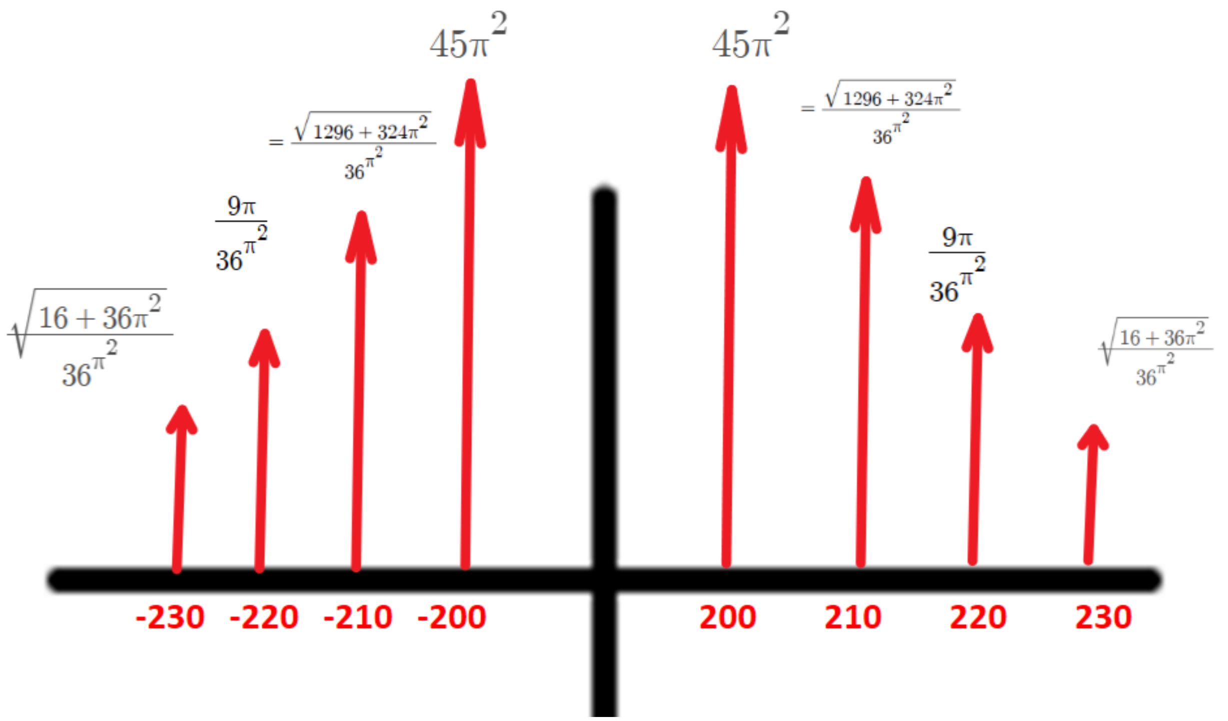


Figure 6 ::spectrum of S(t) single side band

S(t)single side upper =

$$\sqrt{\frac{16 + 36\pi^2}{36^{\pi^2}}}\ cos(2\pi 230t).2 + \frac{9\pi}{36^{\pi^2}}\ cos(2\pi\ 220t).2$$

$$+ \frac{\sqrt{1296 + 324\pi^2}}{36^{\pi^2}}\ cos(2\pi\ 210t).2 + 45\pi^2\ cos(2\pi\ 200t).2$$